

Code-Layout, Readability and Reusability

Date	23 June 2026
Team ID	SWTID-2026-8204
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP-8 style 4-space indentation followed consistently across app.py and model training scripts
2	Proper File Structure	Files and folders are logically organize	Yes	Separated into templates/ (HTML), static/ (CSS), model/ (model.pkl), and app.py (Flask backend)
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g., N, P, K, temperature, humidity, rainfall, predicted_crop used instead of generic names
4	Function / Method Names	Functions are descriptively named	Yes	e.g., predict(), home(), about(), findyourcrop() clearly indicate their role
5	Code Comments	Inline and block comments explain logic	Partial	Key preprocessing and model steps commented; some minor helper sections could use more detail
6	Modular Design	Code is split into reusable functions/modules	Yes	Data preprocessing, model training, and Flask routing kept in separate logical sections/files

7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused imports and exploratory/test code cleaned up before final submission
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation on form fields; try/except handling for model loading can be strengthened

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	model.pkl (trained ML model loader)	Python, Pickle	Loaded in app.py for the /predict route	High

2	Data preprocessing functions (null check, outlier handling, log transform)	Python, Pandas, NumPy	Reused across EDA notebook and pre-deployment data pipeline	High
3	Navigation bar (Home / About / FindYourCrop)	HTML, Jinja2 template	Reused across all three pages via shared template/base layout	High
4	Prediction form component	HTML CSS	Used on FindYourCrop page; reusable for future extensions (e.g., additional crop parameters)	Medium
5	Model evaluation utility (accuracy, precision, recall, F1, confusion matrix)	Python, Scikit-learn	Reused across K-Means, Logistic Regression, and other model comparisons	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear separation of concerns between model, backend, and frontend files
Readability	4	Descriptive naming and consistent style make the code easy to follow
Reusability	4	Core components (model loader, preprocessing functions, templates) are reusable for future enhancements
Documentation / Comments	3	Adequate comments on major logic; could be expanded further for onboarding new developers
Overall Score	4/5	Well-structured, readable, and reasonably reusable codebase suitable for further development and deployment